

MACHINE LEARNING WORKFLOW

1. Data Collection

Data collection is the first step in Machine Learning. In this project, student performance data is collected using study hours, attendance, assignments, and marks. The data is stored in a CSV file called `student_performance.csv`. Good quality data helps the model make accurate predictions. The dataset should contain useful and correct information.

2. Data Cleaning

Data cleaning means checking the dataset for missing values, incorrect data, or duplicate records. Clean data improves the performance of the model. In this project, the dataset is already clean and does not contain missing values. Proper cleaning helps avoid prediction errors.

3. Feature Selection

Feature selection means choosing important input columns for prediction. In this project, `Study_Hours`, `Attendance`, and `Assignments` are selected as features. These features help predict student marks. Selecting relevant features improves model accuracy and efficiency.

4. Train-Test Split

The dataset is divided into training data and testing data. Training data is used to train the model, while testing data is used to evaluate performance. In this project, 80% data is used for training and 20% for testing. This helps check how well the model works on new data.

5. Model Training

Model training is the process where the machine learning algorithm learns patterns from the training data. In this project, Linear Regression is used to train the model. The model learns the relationship between study hours, attendance, assignments, and marks. Training helps the model make future predictions.

6. Testing

Testing is done after training the model. The testing data is given to the model to check prediction accuracy. This step helps measure how well the model performs on unseen data. Good testing results indicate better learning by the model.

7. Evaluation

Evaluation measures the performance of the machine learning model using metrics like Mean Absolute Error (MAE) and R^2 Score. MAE measures prediction error, while R^2 Score shows model accuracy. A low MAE and high R^2 Score indicate good performance.

8. Prediction

Prediction is the final step in the Machine Learning workflow. The trained model predicts marks for new student data based on study hours, attendance, and assignments. Predictions help estimate future outcomes using learned patterns from the dataset.